

Assessing model and item fit in psychometrics

Moving toward up-to-date methods and appropriate cutoff values

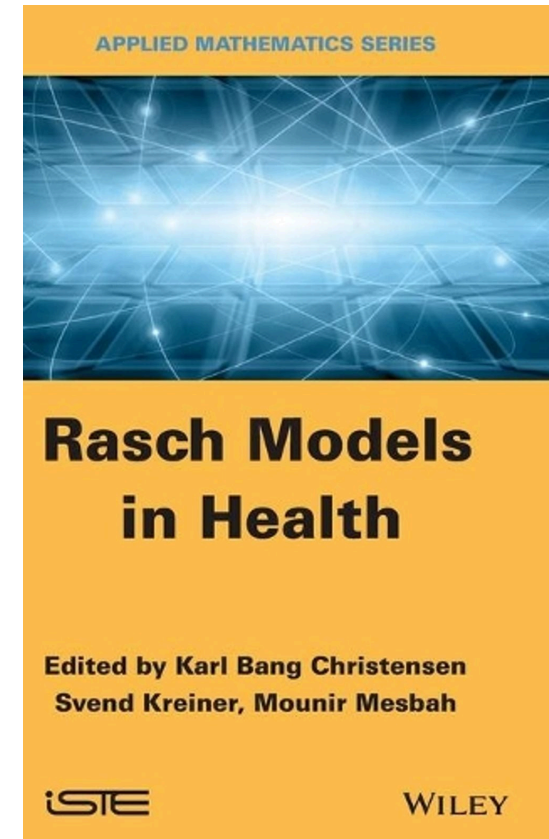
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Challenges of psychometric analyses

- **Evaluating model–data fit** (Christensen et al. 2021)
 - unidimensionality
 - local independence
 - ordered response categories
 - invariance
- **Using adequate methods**
 - with appropriate interpretations



(Christensen et al. 2013)

Before diving in...

Scope

- Examples are primarily related to **Rasch Measurement Theory** (RMT), focusing on unidimensionality and item fit, and local (in)dependence.
- But the concepts and general methods apply to any measurement model.

This will be a non-technical presentation, focused on results and recommendations rather than explanations and details.

 Link to this presentation on GitHub is provided on the last slide

Traditional workflow

- We evaluate how well the data fit the measurement model using metrics such as **item fit statistics**, Q_3 **residuals**, etc.
- But how do we decide what counts as *misfit*?

Critical values — a.k.a. “cutoffs”

- Rule-of-thumb cutoffs dominate practice...
- ...despite strong published evidence that they depend on sample size, number of items, and more.

A better solution → *Simulation-based cutoffs*

Key takeaways

(Sorry for the bluntness...)

- **WRONG:** Item infit/outfit MSQ within 0.5–1.5 or 0.7–1.3 (Wang and Chen 2005)
- **WRONG:** PCA eigenvalue below 2.0 (or < 1.5) (Chou and Wang 2010)
- **WRONG:** Local dependence (Q_3) above 0.3, or 0.2 above mean (Christensen et al. 2017).
- **WRONG:** Confirmatory factor analysis RMSEA/SRMR/CFI/TLI cutoffs based on **Hu & Bentler (1999)** or any other fixed value (Marsh et al. 2004).
- **Conditional** item fit should be used (Christensen et al. 2013; Müller 2020).
- Large sample sizes ($n > 1000$) require special treatment for some metrics (Johansson 2025).
- *Easy to use software that solves these issues is now available.*

Data-adaptive cutoffs

💡 The core idea is simple: instead of comparing a fit statistic against a fixed number, we compare it against the range of values we would expect to see if the data at hand actually fit the model.

- Estimate item & person parameters from **your** sample and items
- Simulations are tailored to mimic your data — and so is the expected distribution of the fit metric
- Compare the **expected** distribution with the **observed** value — *voilà!*

📖 Simulating from model parameters estimated on data = **parametric bootstrap**.

Simulation-based workflow

Setup

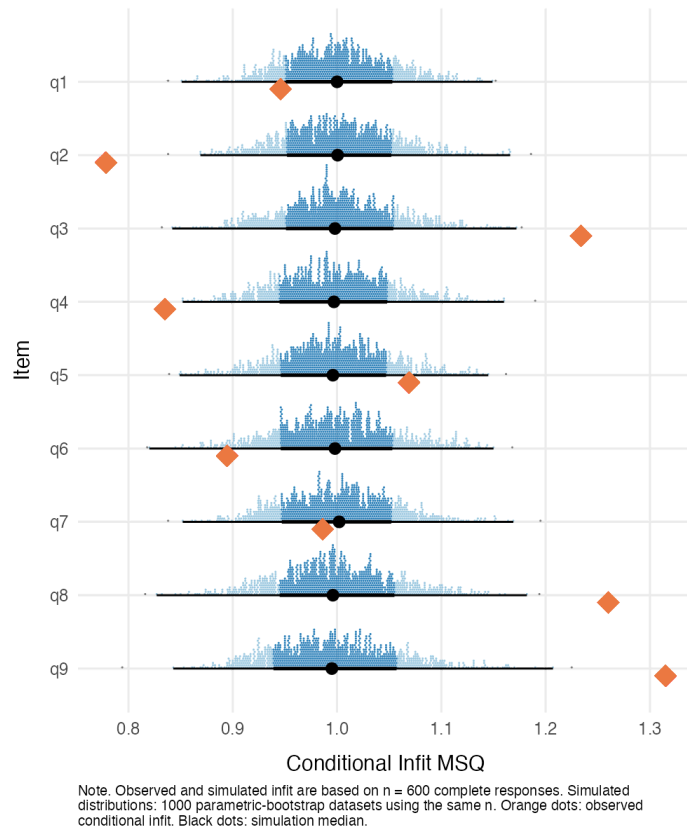
1. Start with the dataset being analyzed
2. Estimate item & person parameters from observed data

Simulation (parametric bootstrap)

3. Resample person parameters (θ) with replacement
4. Using the item parameters, simulate item responses based on θ
5. Estimate the model parameters
6. Calculate the fit metric

 Repeat steps **3–6** many times (~500–1000).

A visual example



Conditional infit MSQ

PHQ-9, $N = 600$

- Blue = simulated (expected) distribution
- Orange ♦ = observed value
- Values out in the tails → flagged as underfit (above) or overfit (below)

How do you get simulation-based cutoffs?

Free, open-source software! Both of these rely on R packages others built.

jamovi overview

Variables

Data

Analyses

Plots

Edit

Exploration

T-Tests

ANOVA

Regression

Frequencies

Factor

easyRasch2jmv

	q1	q2	q3	q4	q5
1	3	3	3	3	
2	0	0	1	1	
3	1	2	3	3	
4	2	3	0	2	
5	3	3	3	3	
6	3	3	1	3	
7	1	1	0	1	
8	3	3	3	3	
9	2	2	2	2	
10	1	0	0	0	
11	2	2	1	2	
12	2	3	3	3	
13	0	2	3	3	
14	1	1	3	2	
15	1	0	1	2	
16	1	1	1	1	
17	1	1	0	1	2
18	2	0	2	1	1
19	2	1	2	1	1

Item-Restscore Correlations

Q3 Residual Correlation Matrix

Conditional Item Infit

Andersen LR-test DIF

Partial Gamma DIF

Item Probability Curves

Dynamic CFA Fit-Index Cutoffs

Targeting Plot

Bootstrap Item-Restscore

Conditional Item Infit (Multiple Imputation)

Reliability

Sum Score to Logit Transformation

PCA of Standardized Rasch Residuals

version 2.7.14

Conditional Item Infit

gender

age

edu

Items

q1

q2

q3

q4

q5

q6

q7

q8

☐ Sort by Infit MSQ (descending)

☒ Compute simulation-based cutoffs

HDCI width

99.9

%

Number of simulation iterations

500

Random seed

42

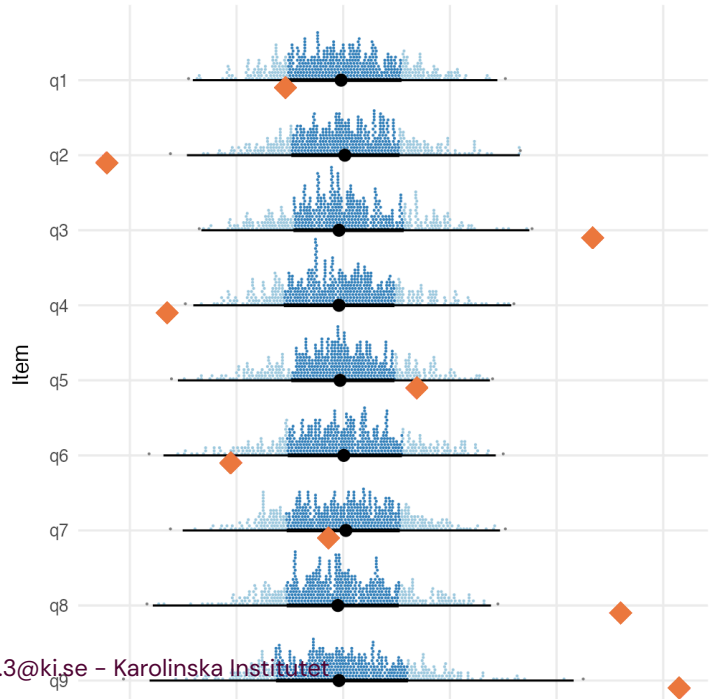
Item	Infit MSQ	Expected range		Flagged	Relative location
		Lower	Upper		
q1	0.946	0.855	1.15		-0.580
q2	0.778	0.838	1.17	TRUE	-0.780
q3	1.234	0.865	1.18	TRUE	-0.850
q4	0.835	0.852	1.16	TRUE	-1.510
q5	1.069	0.839	1.14		-0.600
q6	0.895	0.818	1.15		-0.780
q7	0.986	0.838	1.15		-0.680
q8	1.260	0.816	1.15	TRUE	0.950
q9	1.315	0.794	1.23	TRUE	0.770

Note. Conditional infit MSQ is computed from complete responses only (n = 600 row(s) with no missing values across the selected items).

[3] [4] [5] [6] [7] [8]

Cutoff values based on 500 simulation iterations (99.9% HDCI) drawn from the same n = 600 complete cases.

Simulated Infit MSQ Distribution



Item fit

Two key papers:

- **Müller** (2020) — on conditional item fit.
- **Johansson** (2025) — on power to detect misfit due to multidimensionality, using conditional infit/outfit and item–restscore (Goodman–Kruskal's γ).

⚠ Important

Since unconditional item fit is unreliable with a sample size larger than ~200 (Müller 2020) and the detection of misfit items is generally underpowered with sample sizes below 200 (Johansson 2025), **we should use conditional item infit.**

Conditional item fit MSQ (Müller 2020)

- The paper compares **conditional and unconditional** infit/outfit.
- Unconditional item fit is inconsistent from $n > \sim 200$.
 - ZSTD transformation of MSQ does not help.
- Unconditional/ZSTD *is what is used in all software unless explicitly noted otherwise*.
- For conditional item fit, standard errors are not consistent → we need bootstrap/simulation to interpret results.
- **Limitation:** Requires complete response data.
 - Multiple imputation with chained equations (MICE) can be used.
 - MICE is implemented in jamovi module and **easyRasch2**.

Detecting item misfit (Johansson 2025)

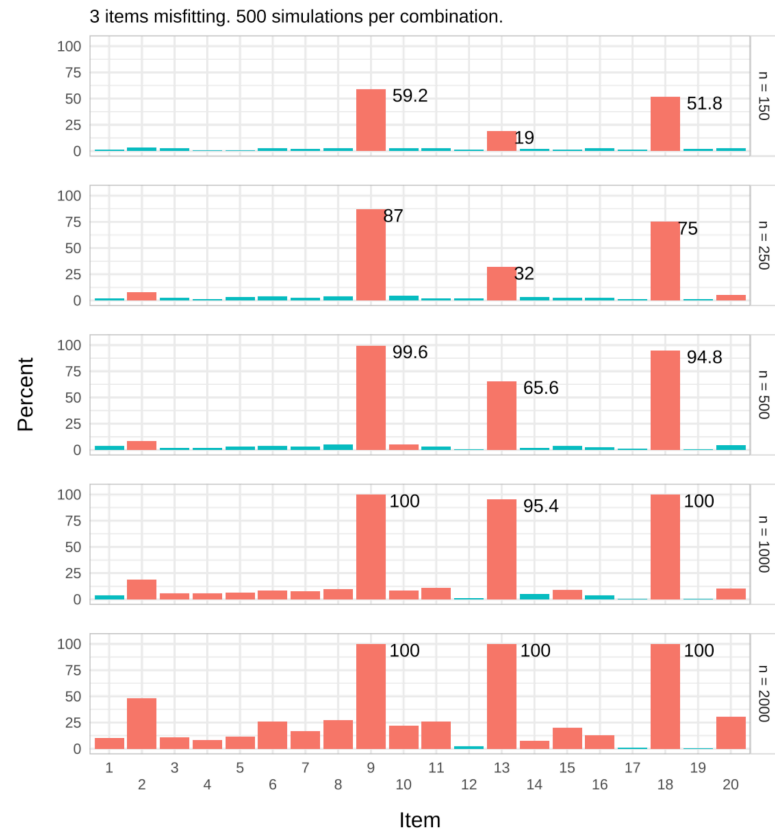
- **Outfit** is much less powerful than conditional **infit** and **item-restscore (GK γ)** in detecting multidimensionality issues (underfit items).
- Conditional infit is slightly more powerful than item-restscore for $n < 500$.
- Off-target items are harder to detect (-2 logits compared to sample mean).
 - ± 1 logit makes little difference.
- Large samples ($n > 1000$) should use item-restscore with non-parametric bootstrap (available in jamovi & **easyRasch2**).

Detecting item misfit 2/3

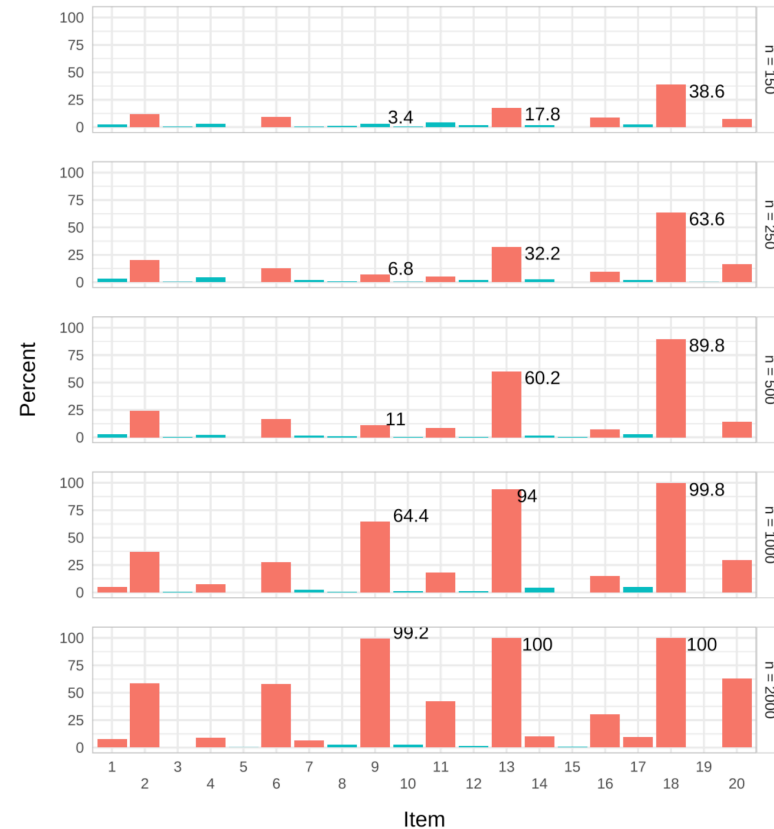
- You will see a plot that shows detection rate (%) of underfit items.
- 20 dichotomous items, of which 3 are underfit due to multidimensionality.
- Targeting for item 9 = same as sample mean
 - item 18 = -1 logits
 - item 13 = -2 logits
- There are panel plots for different sample sizes, from 150 to 2000.

Detecting item misfit 3/3

Conditional infit



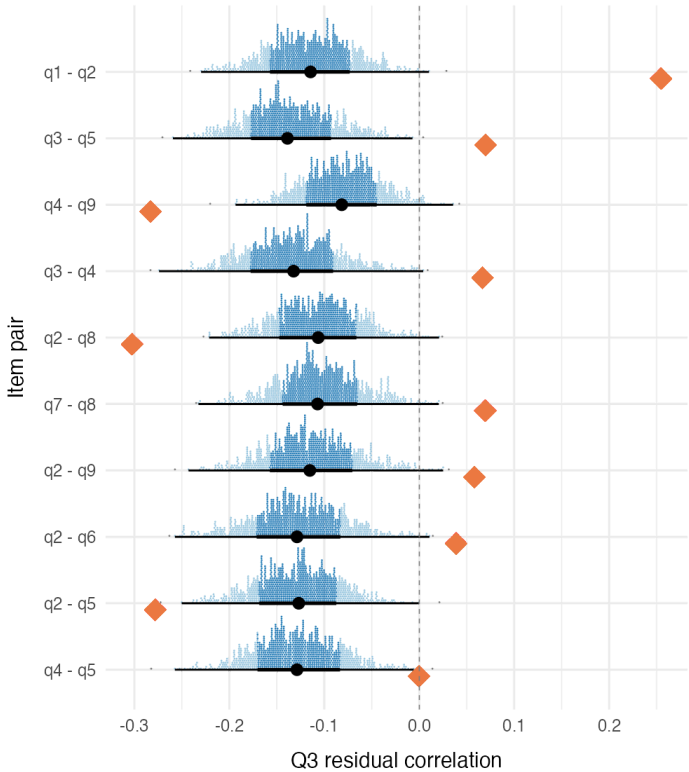
Conditional outfit



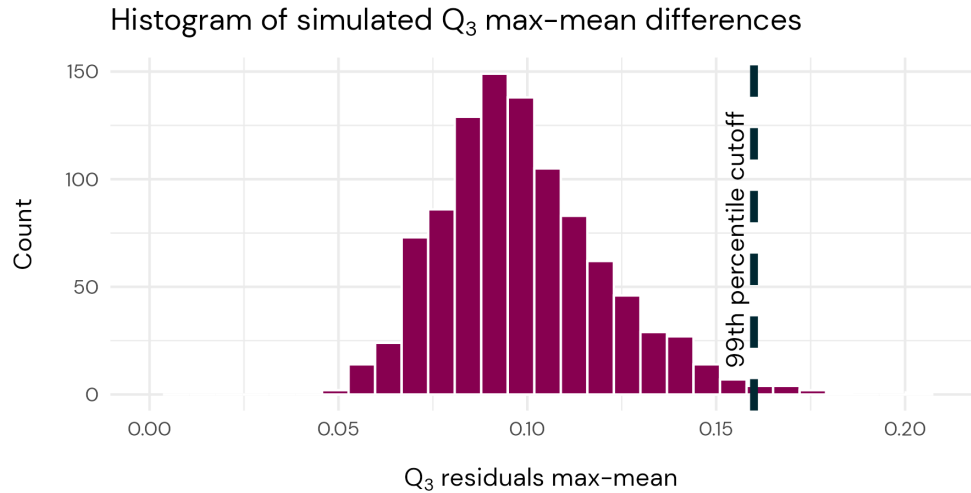
Local dependence: Q_3 residuals

Global cutoff vs. separate item pair cutoffs

	q1	q2	q3	q4	q5	q6	q7	q8	q9	above_cutoff
q1										
q2	0.25									*
q3	-0.19	-0.2								
q4	0	-0.04	0.07							*
q5	-0.21	-0.28	0.07	0						*
q6	-0.18	0.04	-0.2	-0.16	-0.13					
q7	-0.13	-0.2	-0.22	-0.09	-0.14	-0.05				
q8	-0.17	-0.3	-0.17	-0.15	-0.09	-0.17	0.07			*
q9	-0.14	0.06	-0.23	-0.28	-0.24	0.01	-0.18	-0.12		*



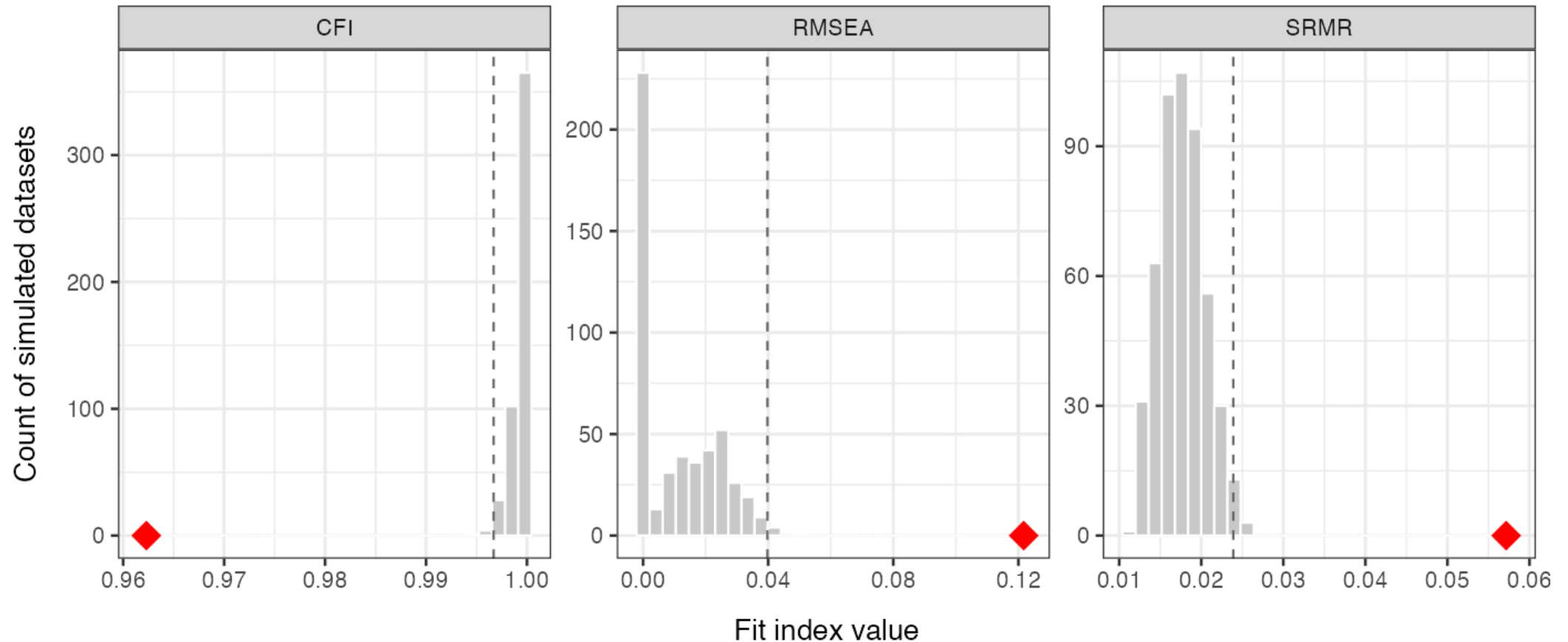
Note. Observed and simulated Q3 are based on n = 600 complete responses. Simulated distributions: 1000 parametric-bootstrap datasets using the same n. Orange diamonds: observed Q3. Black dots: simulation median.



CFA model fit

- Confirmatory factor analysis (CFA) is a test of **dimensionality**.
- Usually judged by several indices — **RMSEA, SRMR, CFI, TLI, etc.**
 - Hu & Bentler (1999): ~150k citations for rule-of-thumb cutoffs.
 - Same issue as other metrics — they don't generalize.
 - Are extra inappropriate for ordinal data.
- **Simulation-based cutoffs should be used** (McNeish and Wolf 2024).
 - R package **dynamic**; but easier with **easyRasch2** and jamovi module.
- **Remember to use a correct estimator for ordinal data!**
 - WLSMV, ULSMV, DWLS, etc. (`lavaan::cfa(data, ordered = TRUE, estimator = "WLSMV")`)

CFA: observed vs. simulated (jamovi)



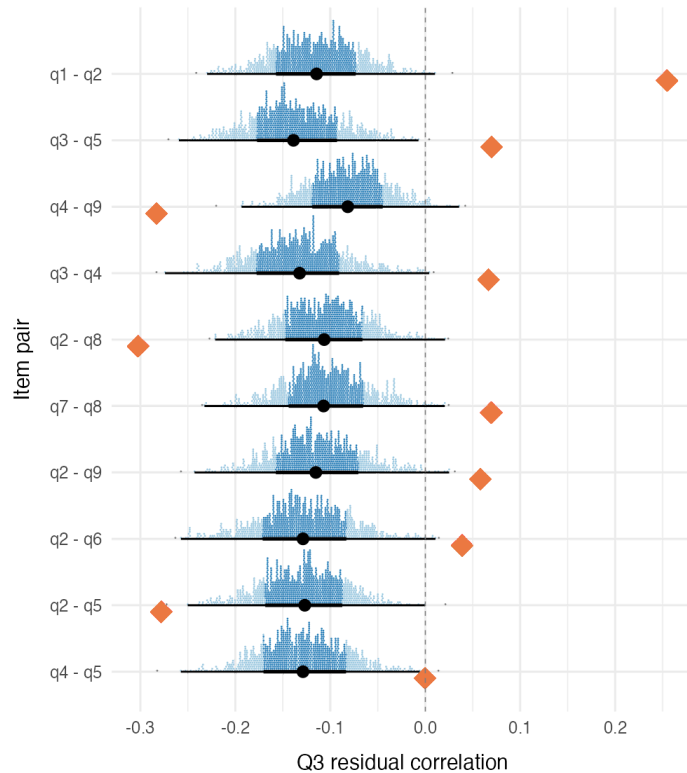
Note. Histograms: 500 parametric-bootstrap datasets simulated under PCM unidimensionality at $n = 600$, refitted with `lavaan::cfa(ordered = TRUE, estimator = "WLSMV")`. Diamond: observed value (red = flagged at the 99th percentile in the unfavourable direction). Dashed line: cutoff (CFI: 1th pct; RMSEA / SRMR: 99th pct).

Where to draw the line?

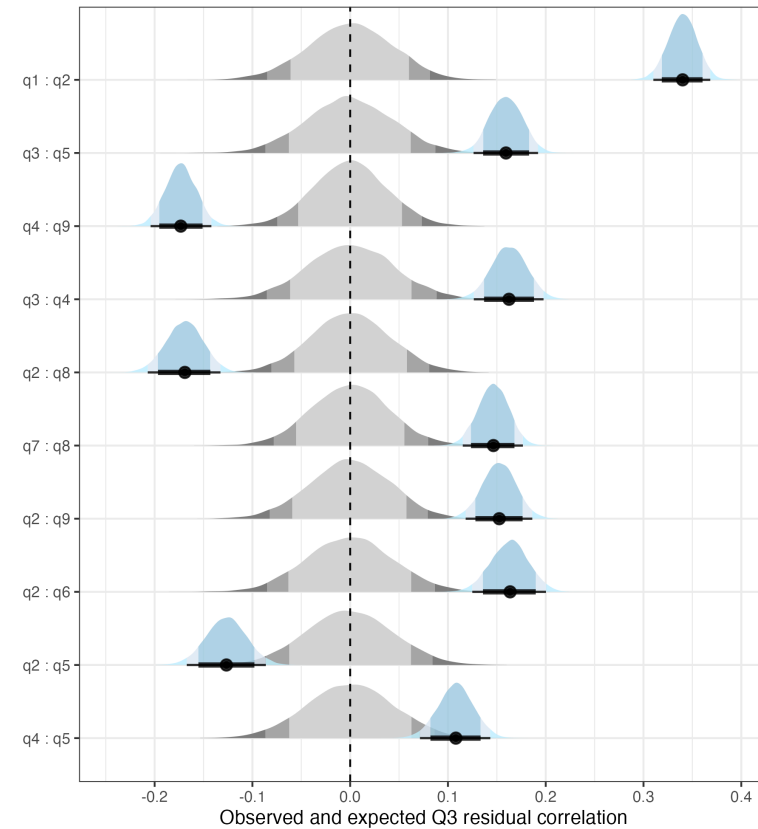
Choosing a cutoff

- Where should we draw the line on the simulated distribution?
 - p -value? Which alpha/correction to use?
 - Distribution-based, 95th percentile? 99th? Fully outside the expected range?
- **Bayesian view:** compare the *expected distribution* to the *posterior distribution* — how much do they overlap?
 - What is the probability that data were generated by the model?
- A similar idea works in a frequentist frame, although we have only one observed value to compare with the simulated distribution of expected values.

Q_3 residuals: simulation vs Bayesian



Note. Observed and simulated Q_3 are based on $n = 600$ complete responses. Simulated distributions: 1000 parametric-bootstrap datasets using the same n . Orange diamonds: observed Q_3 . Black dots: simulation median.



A call to action

- We need to collectively move the norms regarding cutoffs and methods
 - **Please spread the word**
- Use this information when **reviewing and writing papers**
 - In your paper intros, mention limitations for old papers using old methods and/or fixed cutoffs.
 - Don't assume that published papers are correct – most are likely to have errors of various severity
- Editors could have a brief guideline/checklist for authors

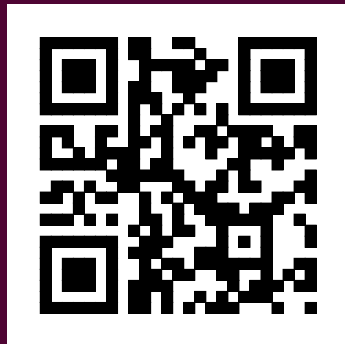
Thank you!

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Slides: <https://pgmj.github.io/SAMC2026/>

Code: <https://github.com/pgmj/SAMC2026/>



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